

DIMHANS

Alcohol Support Application

Project Technical Report

Comprehensive Overview — Architecture, Features & Build Status

1. Executive Summary

DimhansApp is a bilingual (English and Kannada) Flutter-based mobile application developed for the Department of Psychiatry at DIMHANS (Dharwad Institute of Mental Health and Neuro Sciences). The application is purpose-built to deliver comprehensive alcohol addiction support and recovery management to patients and the broader community.

The platform integrates emotional self-awareness tools, crisis management resources, coping skill development modules, and community support features — all backed by real-time Firebase cloud infrastructure. An offline-first architecture ensures the application remains fully functional in low- or no-connectivity environments, a critical requirement given the diverse geographic reach of DIMHANS patients.

Platform	Android (min SDK 23 / Android 6.0 Marshmallow)
Framework	Flutter SDK ^3.8.1 (Dart)
Backend	Firebase (Auth, Firestore, Cloud Storage)
Languages Supported	English and Kannada (bilingual)
Report Date	April 2026

2. System Architecture & Technology Stack

2.1 Core Framework & Language

The application is built on Flutter SDK (targeting version ^3.8.1) using Dart as its primary language. Flutter’s single-codebase paradigm was selected for its performance characteristics and mature widget ecosystem, enabling the team to deliver a polished, platform-adaptive UI without maintaining separate native codebases.

Android is the primary compile target. The minimum SDK level has been set to 23 (Android 6.0 Marshmallow) to satisfy the dependency requirements of the Firebase Authentication library and to ensure broad device coverage across the user base.

2.2 Dependency Inventory

The table below catalogues all major third-party libraries and plugins integrated into DimhansApp, along with their pinned versions and functional roles:

Library / Plugin	Version	Purpose
Flutter SDK	^3.8.1	Core UI Framework
Dart	Latest stable	Primary language
Firebase Auth	^5.7.0	User authentication
Firebase Core	Latest	Cloud backend

Library / Plugin	Version	Purpose
Provider	^6.1.5	State management
Shared Preferences	^2.2.2	Offline storage
HTTP	^1.2.0	REST networking
intl	Latest	Localisation (EN/KN)
url_launcher	Latest	Deep links & web views

2.3 State Management Strategy

The application adopts the Provider pattern (^6.1.5) for state management. Provider was chosen for its clean separation of business logic from the widget tree, making it straightforward to propagate authentication state, mood data, and assessment results across deeply nested screen hierarchies without coupling UI components to data sources.

2.4 Data Persistence & Offline-First Design

DimhansApp employs a hybrid persistence model designed to prioritise continuous availability:

- Online: Firebase Firestore serves as the authoritative cloud datastore, handling real-time synchronisation of mood logs, assessment results, and community contributions.
- Offline: SharedPreferences (^2.2.2) acts as a local buffer for pending user inputs. On connectivity restoration, the app transparently flushes buffered entries to Firestore, ensuring no data is lost.

This approach is particularly important given that many DIMHANS patients may access the application in areas with unreliable mobile data coverage.

3. Application Flow & Screen Topology

Navigation is anchored by a central MainNavigation component that renders a Bottom Navigation Bar, providing thumb-friendly access to all primary modules. The full screen inventory is detailed below.

Module	Key Files	Responsibility
Authentication	auth_wrapper.dart, login_screen.dart, register_screen.dart, email_verification_screen.dart, forgot_password_screen.dart	Entry and credential management
Home	home_screen.dart	Dashboard with prompt links for assessments and mood checks
Learn	learn_screen.dart	Educational portal with public info tabs and specific modules
Help	help_screen.dart	Emergency support and professional contact resources

Module	Key Files	Responsibility
Craving Skills	craving_skills_screen.dart	Actionable coping techniques for trigger moments
Mood Tracker	mood_tracker_screen.dart, mood_history_screen.dart	Mood logging, trigger capture, analytics graphs
Community	community_screen.dart	Peer support section
Assessment	assessment_screen.dart	Self-assessment questionnaire with context-aware feature reveal
About	about_screen.dart	DIMHANS info and personnel profiles
Admin	admin_dashboard_screen.dart, admin_management_screen.dart	Admin-only: content push, data overview, system health

3.1 Authentication & Onboarding Flow

The `auth_wrapper.dart` acts as the root listener, observing Firebase Auth state changes and redirecting users between the authentication stack and the main application shell. The full credential lifecycle is supported:

- New user registration with email/password (`register_screen.dart`).
- Email address verification before full access is granted (`email_verification_screen.dart`).
- Password recovery via the `forgot_password_screen.dart` flow.

This multi-stage verification process ensures the integrity of the user base and protects sensitive health data stored in Firestore.

3.2 Assessment Logic

The `assessment_screen.dart` implements a structured self-assessment questionnaire. Crucially, the flow incorporates conditional branching: if a user indicates active alcohol consumption, additional context-aware tracking features — most notably the Mood Tracker — are surfaced. This progressive disclosure approach prevents overwhelming new users and ensures feature relevance.

3.3 Admin Workflows

Administrative functionality is fully segregated from public-facing screens. The `admin_dashboard_screen.dart` and `admin_management_screen.dart` are accessible only to accounts with administrator privileges, enforced at the service layer by `admin_service.dart`. Administrators can:

- Push new educational content to the Learn portal.
- Review aggregated user data and system health metrics.
- Monitor overall platform usage without exposing any individual user data to other users.

4. Feature Deep-Dive: Mood Tracker

The Mood Tracker is the most clinically significant feature in DimhansApp, designed in direct alignment with evidence-based addiction recovery principles. It integrates tightly with Assessment results to ensure only users who have indicated alcohol use are prompted to log mood data.

4.1 Input Model

- Mood Scale: Users log their current emotional state on a standardised 5-point scale.
- Trigger Capture: A structured selection mechanism records the cause behind each mood entry, covering categories such as Financial Problems, Social Pressure, and Boredom — common antecedents of alcohol craving.
- Coping Strategy Tracking: The app records which coping technique (e.g. Meditation, Nature Walks, Breathing Exercises) was employed, enabling longitudinal analysis of strategy efficacy.

4.2 Offline-First Data Flow

Mood entries adhere to the application-wide offline-first model. When network connectivity is unavailable, entries are serialised to SharedPreferences. A background sync listener monitors connectivity and automatically pushes pending entries to Firebase Firestore upon reconnection, guaranteeing data integrity with no manual user intervention required.

4.3 Analytics & Visualisation

DimhansApp renders bespoke analytics charts to help users and clinicians understand recovery trajectories:

- 7-day, 30-day, and 90-day mood trend graphs.
- Frequency analysis mapping specific coping strategies against self-reported success rates.
- Trigger pattern visualisation to identify recurring high-risk situations.

These visualisations are generated entirely on-device from Firestore-synced data, ensuring up-to-date insights without requiring a dedicated analytics backend.

5. Recent Build & Infrastructure Updates

The following engineering changes were executed during the most recent release cycle to address Android Gradle compatibility issues and harden the build pipeline:

Update	Detail
Plugin Loader Decoupling	Resolved Gradle classpath clash via flutter-plugin-loader in settings.gradle.kts; removed hardcoded version constraint
MinSDK Alignment	Raised minimum SDK to 23 (Android 6.0 Marshmallow) to satisfy firebase-auth:23.2.1 internal dependency requirements
Release APK Output	Successfully compiled release APK at build/outputs/flutter-apk/app-release.apk
Repository Sync	All configuration changes committed and pushed to the main GitHub repository branch

5.1 Gradle Plugin Loader Decoupling

A plugin classpath conflict was identified in the legacy Gradle configuration, caused by a hardcoded version "1.0.0" constraint within app/build.gradle.kts. This was resolved by migrating to Flutter's dynamic flutter-plugin-loader mechanism, configured inside settings.gradle.kts. This approach decouples plugin

resolution from the application build script, enabling seamless updates to Flutter tooling without manual version pinning.

5.2 Minimum SDK Alignment

The firebase-auth package at version 23.2.1 enforces a minimum Android SDK of 23 via its internal dependency tree. The project's minimum SDK target was updated accordingly. This change is backwards-compatible with the target user base and unlocks future use of modern Android APIs available from API 23 onward.

5.3 Release Build

A successful release APK was compiled and is available at the following path within the repository:

```
build/outputs/flutter-apk/app-release.apk
```

All build configuration changes have been committed to and pushed on the main branch of the GitHub repository.

6. Bilingual Localisation

Kannada localisation is a foundational requirement of DimhansApp, reflecting the primary language of the patient population served by DIMHANS in northern Karnataka. The intl package is employed to manage:

- Form labels and input field placeholders.
- Screen headers and navigation labels.
- UI state messages (e.g. loading indicators, error messages, success confirmations).
- Date and number formatting appropriate for the locale.

Developer Requirement: All developers contributing UI elements to DimhansApp are required to maintain parallel translations in both English and Kannada. Any new string introduced without a corresponding Kannada translation is considered an incomplete implementation and must be resolved before merge.

7. Summary & Next Steps

DimhansApp represents a clinically grounded, technically robust mobile platform for alcohol addiction support. The current build delivers a complete feature set spanning authentication, self-assessment, mood tracking, psychoeducation, and crisis resources — all within a bilingual, offline-capable shell.

The recent Gradle and Firebase compatibility updates ensure a stable, forward-compatible Android build pipeline. The release APK is ready for internal testing and staged distribution.

Recommended Next Actions

- Conduct structured UAT (User Acceptance Testing) with a cohort of DIMHANS clinical staff and patient representatives.
- Validate Kannada translations with native Kannada speakers for medical accuracy and cultural appropriateness.

- Implement Firebase Remote Config to enable clinician-controlled content updates without requiring app store releases.
 - Explore iOS build target as a future platform extension.
 - Integrate push notifications for medication reminders and motivational check-ins.
 - Commission a security audit of Firebase rules and data access controls prior to public release.
-